

# Lecture 14: Linear Constrained Optimization

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SDU

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DEPARTMENT OF MATHEMATICS  
AND COMPUTER SCIENCE

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# 1 Introduction to this course

- Continuous multivariate optimization
  - Single-valued, single-variate
    - $f : \mathbb{R} \rightarrow \mathbb{R}$
  - Single-valued, multivariate
    - $f : \mathbb{R}^n \rightarrow \mathbb{R}$
  - Vector-valued, multivariate
    - $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$
    - Not correiculum
- Discrete
  - $f : \mathbb{Z}_0^+ \rightarrow \mathbb{R}$

## Contents of the course

- Introduction, univariate problems
- Multivariate Problems, Gradient-Based Methods
- Derivative free methods
- Optimization for Machine Learning
- Constrained Optimization, Linear Programming
- Sampling methods
- Discrete Optimization and Heuristics

Communication Means GitHub → <https://ai-505.github.io> ItsLearning Write to Professor Marco (marco@imada.sdu.dk) Alternatively to instructor Sai (sgnagarajan@imada.sdu.dk)

## Assessments

- Mandatory assignments in groups of 2:
  - Assignment 1
  - Assignment 2
  - Oral exam on June 29-30, 2026

Oral exam consists of questions based on the assignments + can be extended to cover other parts of the curriculum

Final grade: primarily based on the assignments but the oral exam may adjust the grade up or down by one grade level.

## Exercise sessions

- + are done before class
- \* are done in class
- unmarked are for self study

Note that are good examples of assignment questions.

Schedule for the year ![[Pasted image 20260203122456.png]]

Objectives to optimize

- Efficiency
- Safety
- Accuracy

Constraints

- Cost
- Weights
- Structural integrity

Challenges

- High-Dimensional Search Spaces
- Multiple Competing Objectives
- Model Uncertainty

Model, Representation, Implementation

- Model: mathematical object in some class  $M$
- Representation: An object of an abstract data type  $R$  used to store the model
- Implementation: An object of a concrete type used to store the model.
- Any object from the real world might have different models.
- Any model might have several representations (exact).
- Any representation might have different implementations (exact).

We will focus on the algorithmic aspects of optimization that arise after the problem has been properly formulated

## 2 The optimization process

Multiple different designs for optimization a process

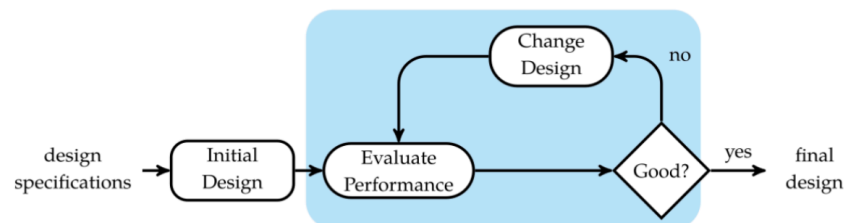


Figure 1: This is very simple, but it illustrates **incrementally improving a design** until it can no longer be improved or until the budgeted time or cost has been reached.

### 2.1 White/glass box

Evaluating an analytical expression. This is expressible mathematically.

## 2.2 Black box

Running physical experiments, e.g. wind tunnel tests. These are more difficult to express mathematically, and will probably not result in a formula/mathematical expression.

Note that computer simulation is also a black box, but it might be less computationally expensive.

## 2.3 computer simulations

## 3 Basic optimization problem

minimize  $f(x)$  w.r.t.  $x$  where  $x \in \mathcal{X}$

this is formulated as such

$$x^* = \arg \min_{x \in \mathcal{X}} f(x)$$

Here

- Feasible Set  $\mathcal{X}$
- Design Point  $x$
- Design Variables
- Objective Function:  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  (scalar-valued function)
- Minimizer

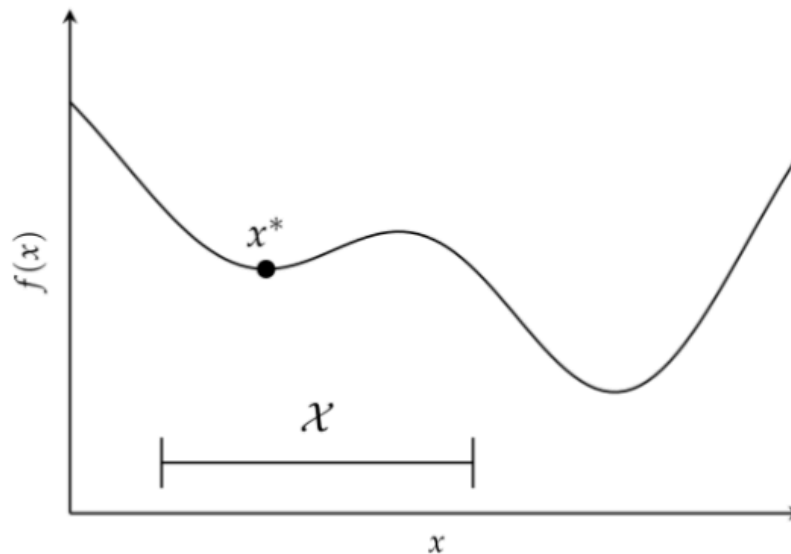


Figure 2: Example of a basic optimization problem.

Remember that there is only one global minimum, but there might be many minimizers

Also

$$\arg \max_{x \in \mathcal{X}} f(x) \equiv \arg \min_{x \in \mathcal{X}} -f(x)$$

### 3.1 Constraints

Take the following example

$$\begin{array}{ll}\underset{x_1, x_2}{\text{minimize}} & f(x_1, x_2) \\ \text{subject to} & x_1 \geq 0 \\ & x_2 \geq 0 \\ & x_1 + x_2 \leq 1\end{array}$$

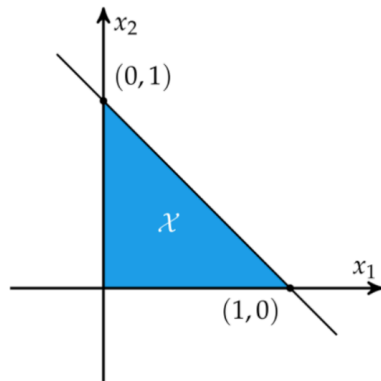


Figure 3:

$$\begin{array}{ll}\min_x & f(x) \\ \text{s.t.} & x_1 \geq 0 \\ & x_2 \geq 0 \\ & x_1 + x_2 \leq 1\end{array}$$

$$x \in \mathbb{R}^2$$

$$x^* = \arg \min_{x_1, x_2} f(x_1, x_2) \quad \text{where } 0 \leq x_1, x_2, \quad 1 \geq x_1 + x_2$$

### 4 Conditions for local minima

Univariate objective functions  $f'(x^*) = 0$ , first-order necessary condition (FONC)  $f''(x^*) \geq 0$ , second-order necessary condition (SONC)

Multivariate objective functions  $\nabla f(x^*) = 0$ , first-order necessary condition (FONC)  $\nabla^2 f(x^*) \geq 0$ , second-order necessary condition (SONC)

Note that here  $\nabla^2 f(x^*)$  is the hessian matrix

## 5 Taylor Expansion

Because of

$$f(x+h) = f(x) + \int_0^h f'(x+a) \, da$$

then by 

$$f(x+h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} h^n$$

Then for multiple dimensions

$$f(x) = f(a) + \nabla f(a)^T (x-a) + \frac{1}{2} (x-a)^T \nabla^2 f(a) (x-a) + \dots$$

## 6 Benchmark functions

Classic benchmark functions (Rastrigin, Rosenbrock, Ackley, Sphere, etc.) have well-studied shapes. We can use this to test whether our optimizer is good or not

Benchmarks tell you whether your optimizer is competent enough to even deserve being tested on a real ML problem.

## 7 Problem clarification

- Univariate  $f : \mathbb{R} \rightarrow \mathbb{R}$  vs Multivariate  $f : \mathbb{R}^n \rightarrow \mathbb{R}$
- Real-valued  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  vs vector functions  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
- Linear vs Nonlinear
- Nonlinear: Convex vs Nonconvex, unimodal vs multimodal
- Constrained vs unconstrained
- Smooth (differentiable) vs non smooth (non differentiable)
- Deterministic vs Uncertain
- Continuous vs Discrete